

Drugstore AI Assistant: RAG and Agentic AI System

1. Introduction

In this project we developed a Drugstore AI Assistant that combines a Large Language Model (LLM), Retrieval-Augmented Generation (RAG), and agentic tool use. The system allows users to ask natural-language questions about a database of 100 drugstore products and receive answers grounded in the available data.

The database contains product name, category, brand, package size, primary use, and price in ILS. The assistant supports both quantitative questions, such as “Which products cost less than 20 ILS?”, and qualitative questions, such as “What products can help with a headache?”

Two Qwen models were compared: **Qwen2.5-1.5B-Instruct** and **Qwen2.5-7B-Instruct**, allowing us to examine how model size affects tool selection and answer quality.

2. Dataset and Preprocessing

The dataset was provided as an Excel file containing 100 drugstore products. Since prices were stored as text, a numeric `_numeric_price` column was created for calculations such as averages, minimums, maximums, and price filtering.

Each product was also converted into a text document by combining its available fields. These documents were used for retrieval.

The system therefore uses two complementary approaches: structured DataFrame queries for numerical questions and retrieval for qualitative questions.

3. Model and RAG Architecture

The main model is **Qwen2.5-7B-Instruct**, loaded using 4-bit BitsAndBytes quantization to reduce memory usage and make inference practical in Google Colab.

The retrieval system combines:

- **FAISS + SentenceTransformers (all-MiniLM-L6-v2)** for semantic similarity.
- **BM25** for keyword-based retrieval.

A router decides which approach is appropriate. Quantitative questions are sent to `query_table`, while qualitative and medical-related questions are sent to `semantic_search`.

For example, “How many products cost less than 20 ILS?” is calculated directly from the DataFrame, while “What products are used for cleansing?” requires semantic retrieval.

This combination helps the system handle both exact numerical questions and less predictable natural-language queries.

4. Agentic AI

The RAG pipeline was extended into an agentic architecture. Instead of following one fixed process, the LLM can select tools based on the question and use their results to decide what to do next.

The agent has access to `search_products`, `by_category`, `semantic_search`, `query_table`, `calculator`, and `final_answer`.

The overall workflow is:

User Question → Agent → Tool Selection → Tool Result → Agent → Final Answer

For example, a request for affordable pain-relief products can involve identifying the relevant products, applying a price constraint, and then presenting the retrieved results.

This makes the system more flexible than using a single retrieval operation for every question.

5. Model Comparison

The following experiment used the same router prompt with both Qwen models. The complete prompt and outputs are included below.

Question	Qwen/Qwen2.5-1.5B-Instruct	Qwen/Qwen2.5-7B-Instruct	True Answer *
I have a headache, what medication do you have for me?	Nurofen 200 mg Tablets - Ibuprofen pain/fever relief - ₪22	For headache relief, you can choose from the following medications: - Nurofen 200 mg Tablets: ₪22 - Advil 200 mg Tablets: ₪18 - Acamol 500 mg Tablets: ₪14 - Tylenol Extra Strength: ₪24 - Aspirin 100 mg: ₪12 These options contain ibuprofen, paracetamol, or aspirin, which are commonly used for pain and fever relief.	Pain Relief medications: Advil 200mg Tablets 18₪ Nurofen 200mg Tablets 22₪ Acamol 500 mg Tablets: ₪14 Tylenol Extra Strength: ₪24 Aspirin 100 mg: ₪12 *Pain relief Voltaren Emulgel (defined as pain relief medication) was correctly omitted.

Question	Qwen/Qwen2.5-1.5B-Instruct	Qwen/Qwen2.5-7B-Instruct	True Answer *
What products are used for cleansing?	The database does not contain any information about products used for cleansing.	- Dove Deeply Nourishing Body Wash (Body cleanser, 500 ml, ₱18) - Garnier Micellar Water (Makeup-removing cleanser, 400 ml, ₱25) - Hydrogen Peroxide 3% (Topical cleansing solution, 250 ml, ₱10) - La Roche-Posay Toleriane Cleanser (Facial cleanser, 200 ml, ₱70) - CeraVe Foaming Cleanser (Facial cleanser, 236 ml, ₱55) - Cetaphil Gentle Skin Cleanser (Facial cleanser, 236 ml, ₱55) - Sterimar Nasal Spray (Nasal cleansing, 50 ml, ₱32)	<u>Cleanser in the Response</u> -Hydrogen Peroxide 3% -Dove Deeply Nourishing Body Wash -CeraVe Foaming Cleanser -La Roche-Posay Toleriane Cleanser -Cetaphil Gentle Skin Cleanser -Garnier Micellar Water -Sterimar Nasal Spray <u>Missing in Response</u> -Dove Original Beauty Bar (Moisturizing cleansing bar) <u>Correct 7/8</u>

Question	Qwen/Qwen2.5-1.5B-Instruct	Qwen/Qwen2.5-7B-Instruct	True Answer *
How much does the most expensive product cost?	The most expensive product in the database is an Omron Blood Pressure Monitor, which costs ₪220.	The most expensive product costs ₪220, which is the typical price for the Omron Blood Pressure Monitor in ILS.	Correct
Which products cost less than 20 ILS?	The following items have a price of less than 20 ILS: 1. Acamol 500 mg Tablets - ₪14 2. Advil 200 mg Tablets - ₪18 3. Band-Aid Flexible Fabric - ₪18 4. Blistex Medicated Lip Balm - ₪16 5. Burt's Bees Beeswax Lip Balm - ₪18 6. Colgate Max White - ₪15 7. Colgate Total Original - ₪12 8. Colgate Pro-Expert - ₪15 9. Dove Original Beauty Bar -	The products that cost less than 20 ILS are: - Advil 200 mg Tablets: ₪18 - Acamol 500 mg Tablets: ₪14 - Aspirin 100 mg: ₪12 - Rennie Peppermint: ₪18 - Band-Aid Flexible Fabric: ₪18 - Hansaplast Classic: ₪16 - 3M Micropore Tape: ₪15 - Hydrogen Peroxide 3%: ₪10 - Isopropyl Alcohol 70%: ₪12 - Savlon Antiseptic Cream: ₪18	Correct 13 first products Missing 16 products

Question	Qwen/Qwen2.5-1.5B-Instruct	Qwen/Qwen2.5-7B-Instruct	True Answer *
	₺8 10. Dove Original Antiperspirant - ₺15 11. Dove Daily Moisture Shampoo - ₺18 12. Garnier Fructis Conditioner - ₺	- Colgate Total Original: ₺12 - Colgate Max White: ₺15 - Oral-B Pro-Expert: ₺15	
How many total products are in the database?	There are 100 total products in the database.	There are 100 total products in the database.	Correct
What is the price of Nurofen?	There are no records in the database related to the product "Nurofen". Therefore, there is no information available about its price.	The price of Nurofen 200 mg Tablets is ₺22.	Correct
What is the average price of an item?	The average price of an item is 36.09	The average price of an item is \$36.09.	The average is correct the sign is wrong

*Correct answer compare True to Qwen/Qwen2.5-7B-Instruct Model

6. Results and Analysis

The comparison shows a clear advantage for the **7B model on semantic understanding**.

Both models handled simple numerical questions well. For example, both correctly identified that the database contains 100 products and that the most expensive product costs ₺220.

However, the differences became more visible when the question required interpretation. For the headache query, the 1.5B model returned only Nurofen, while the 7B model identified several relevant products. More importantly, the 1.5B model completely failed on the cleansing query and claimed that the database contained no relevant information. The 7B model successfully retrieved several cleansing-related products.

The Nurofen price query provides another clear example: the 1.5B model incorrectly reported that no matching product existed, while the 7B model correctly returned ₺22.

There was also a small output-formatting problem. Both models calculated the average price correctly as **36.09**, but the 7B model displayed it using a dollar sign rather than ILS. This shows that correct retrieval and calculation do not always guarantee correct final formatting.

Overall, the experiment suggests that the larger model provides stronger semantic understanding and more reliable interaction with the retrieval system.

7. Limitations and Future Work

The system is limited by the information contained in the database. If a product or use is not represented, the assistant cannot reliably provide information about it.

Another important limitation is that the current implementation **does not perform LoRA fine-tuning**. The model is loaded with 4-bit quantization for efficient inference, but its parameters are not trained on the drugstore dataset. Therefore, the project demonstrates RAG, agentic AI, and model comparison, while actual parameter-efficient fine-tuning remains future work.

A natural next step would be to create a labeled training set containing user questions, correct tools, retrieved products, and expected answers, and then apply LoRA. Different training hyperparameters could be tested to measure their effect.

Future evaluation could also use a larger labeled test set and metrics. Output validation could ensure that mentioned products exist in the database and that prices are consistently formatted in ILS.

8. Conclusion

The Drugstore AI Assistant demonstrates how an LLM can be combined with structured data, semantic retrieval, BM25, and agentic tools to create a practical domain-specific assistant.

The experiments showed that both models handled simple numerical tasks successfully, but **Qwen2.5-7B-Instruct performed substantially better on semantic and product-related questions**. It was particularly effective when the user's wording did not directly match the terminology in the database.

The agentic architecture further improves flexibility by allowing the system to select different tools for different types of questions.